MARKING SCHEME

Model Paper — 05

Third Term Test

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Note (for teachers):

- Part I – A: 2 marks per question (generally 1 mark for the correct method and 1 mark for the correct final answer; where there is no working, 1 mark for a correct key step and 1 for the answer). Total 30 marks.
- Part I – B: 10 marks per question. Total 20 marks. Part II: 10 marks per question; candidates answer any **five**. Total 50 marks. The marks for each step are shown in the right-hand column.
- Award full marks for correct alternative methods. For values read from graphs, drawings and measurements, accept a reasonable range as shown.

Part I – A — Answers

1. $T_n = a + (n - 1)d = 50 + 9 \times (-4) = 50 - 36$

Final answer: 14

(2 marks)

2. $11010_2 = 26$, $1011_2 = 11$; $26 - 11 = 15 = 1111_2$ (or direct subtraction in base two: 11010 − 01011 = 01111)

Final answer: 1111_2

(2 marks)

3. $0.375 = \frac{375}{1000} = \frac{3}{8}$

Final answer: $\frac{3}{8}$

(2 marks)

4. $\frac{42}{60} \times 100$

Final answer: 70%

(2 marks)

5. $(a + 3b)(a - 3b) = a^2 - (3b)^2$

Final answer: $a^2 - 9b^2$

(2 marks)

6. $(a^2b)^3 = a^6b^3$; $a^6b^3 \div a^4b = a^{6-4}b^{3-1}$

Final answer: a^2b^2

(2 marks)

7. 47 850 lies between 47 000 and 48 000; the hundreds digit is 8 (≥ 5), so round up

Final answer: 48 000

(2 marks)

8. $4 + 11 = 2x + 3x \Rightarrow 15 = 5x$ (check: $4 - 9 = -5$ and $6 - 11 = -5$)

Final answer: $x = 3$

(2 marks)

9. Points at a fixed distance from AB lie on both sides of it: one line parallel to AB, 3 cm from it on each side

Final answer: a pair of straight lines parallel to AB, one on each side of AB, each 3 cm from AB

(2 marks)

10. $\hat{A} + \hat{B} = 90° \Rightarrow 2\hat{B} + \hat{B} = 90° \Rightarrow 3\hat{B} = 90°$

Final answer: $\hat{B} = 30°$

(2 marks)

11. $2\pi r = 132 \Rightarrow 2 \times \frac{22}{7} \times r = 132 \Rightarrow r = 132 \times \frac{7}{44}$

Final answer: 21 cm

(2 marks)

12. $9^2 + 12^2 = 81 + 144 = 225 = 15^2$ (the square on the longest side equals the sum of the squares on the other two)

Final answer: Yes, since $9^2 + 12^2 = 15^2$ (converse of Pythagoras' theorem)

(2 marks)

13. $2 < 2x \leq 10 \Rightarrow 1 < x \leq 5$

Final answer: $x = 2, 3, 4, 5$

(2 marks)

14. $n(A') = n(U) - n(A) = 30 - 18$

Final answer: 12

(2 marks)

15. $C = 2\pi r \Rightarrow r = \frac{C}{2\pi}$

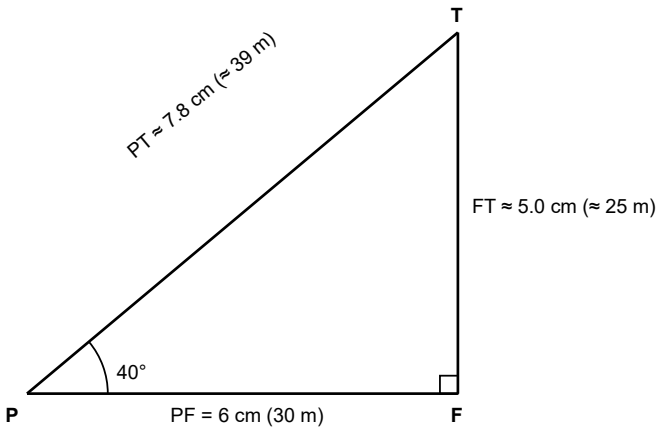
Final answer: $r = \frac{C}{2\pi}$

(2 marks)

Part I – B — Answers and Mark Allocation

Question 1

(i) Rough sketch	
Right-angled triangle PFT with the right angle at F (F directly below T, P and F on level ground), P, F and T labelled	**1**
PF = 30 m and the angle of elevation 40° at P marked	**1**
(ii) Scale diagram (1 cm = 5 m)	
PF = 30 ÷ 5 = 6 cm drawn as the base	**1**
40° angle drawn at P and a perpendicular at F (right angle) drawn to meet the 40° line at T	**1**

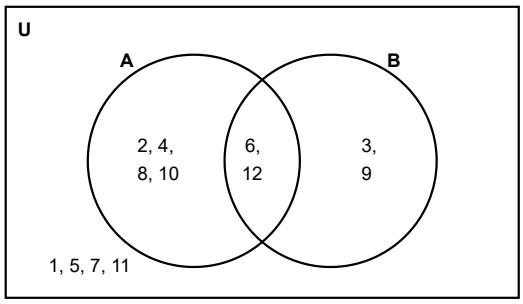


A correct scale diagram (shown reduced in size)

(iii) Height of the tower	
FT measured ≈ 5.0 cm (theoretical: 6 tan 40° = 5.03 cm); accept 4.8 – 5.2 cm	**1**
Height = 5.0 × 5 ≈ 25 m (theoretical 30 tan 40° = 25.2 m; accept 24 – 26 m)	**1**
(iv) Distance PT	
PT measured ≈ 7.8 cm (theoretical: 6 ÷ cos 40° = 7.83 cm); accept 7.6 – 8.0 cm	**1**
PT = 7.8 × 5 ≈ 39 m (theoretical 39.2 m; accept 38 – 40 m)	**1**
(v) Angle of depression	
40°	**1**
Reason: the horizontal line through T is parallel to PF, so the angle of depression of P from T equals the angle of elevation of T from P (alternate angles)	**1**

Total: 10

Question 2

(i)	
A = {2, 4, 6, 8, 10, 12}	**1**
B = {3, 6, 9, 12}	**1**
(ii) Venn diagram	
Rectangle U with two overlapping circles A and B correctly labelled	**1**
All 12 elements placed correctly: only A: 2, 4, 8, 10 ; both: 6, 12 ; only B: 3, 9 ; outside both: 1, 5, 7, 11	**1**
	
(iii)	
A ∩ B = {6, 12}	**1**
A ∪ B = {2, 3, 4, 6, 8, 9, 10, 12}	**1**
(iv)	
(A ∪ B)′ = elements of U not in A ∪ B	**1**
= {1, 5, 7, 11}	**1**
(v)	
A ∩ B′ = {2, 4, 8, 10}, so n(A ∩ B′) = 4	**1**
Description: the numbers in U that are multiples of 2 but not multiples of 3 (elements of A that are not in B)	**1**

Total: 10

Part II — Answers and Mark Allocation

Candidates answer any five questions. (10 marks each)

Question 1

(a)	
$2\frac{1}{2} = \frac{5}{2}, 1\frac{1}{3} = \frac{4}{3}, 1\frac{1}{5} = \frac{6}{5}$	**1**
$\frac{5}{2} - \frac{4}{3} = \frac{15-8}{6}$	**1**
$= \frac{7}{6}$; $\frac{7}{6} \times \frac{6}{5}$	**1**
$= \frac{7}{5} = 1\frac{2}{5}$	**1**
(b)	
(i) $\frac{3}{4} - \frac{3}{8} = \frac{6-3}{8}$	**1**
$= \frac{3}{8}$ of the tank	**1**
(ii) $\frac{3}{8}$ of the capacity = 150 litres $\Rightarrow$ capacity = $150 \div \frac{3}{8} = 150 \times \frac{8}{3}$	**1**
= 400 litres	**1**
(iii) Initial amount = $\frac{3}{8} \times 400$	**1**
= 150 litres (check: $\frac{3}{4} \times 400 = 300 = 150 + 150$)	**1**

Total: 10

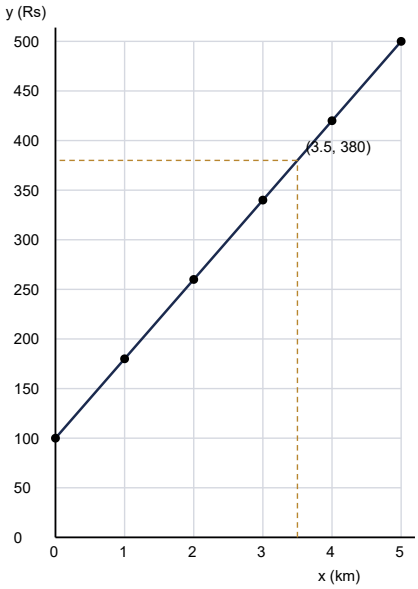
Question 2

(a)	
$(2x^3)^2 = 4x^6$	**1**
$4x^6 \times 3x^{-4} = 12x^{6+(-4)}$	**1**
$= 12x^2$	**1**
(b)	
$125 = 5^3$, so $5^{2x-1} = 5^3$	**1**
$2x - 1 = 3 \Rightarrow 2x = 4$	**1**
$x = 2$ (check: $5^3 = 125$)	**1**
(c)	
$(6.4 \div 1.6) \times 10^{6-2} = 4 \times 10^4$	**1**
$= 4 \times 10^4$ (= 40 000), already in scientific notation	**1**
(d)	
(i) 3.14	**1**
(ii) 3.1416	**1**

Total: 10

Question 3

(i)	
$x = 1 \Rightarrow y = 180$; $x = 3 \Rightarrow y = 340$	**1**
$x = 4 \Rightarrow y = 420$	**1**
(ii) Graph (1 cm = 1 km on the x-axis, 1 cm = Rs 50 on the y-axis)	
Axes labelled and correct scales used	**1**
Points (0, 100), (1, 180), (2, 260), (3, 340), (4, 420), (5, 500) plotted correctly	**1**
Points joined by a single straight line (a ruler is used)	**1**



A correct graph (shown reduced in size)

(iii)	
A line drawn from x = 3.5 up to the graph and across to the y-axis	**1**
Fare = Rs 380 ($80 \times 3.5 + 100 = 380$; accept Rs 375 – 385)	**1**
(iv)	
The gradient = 80 = the charge per kilometre (Rs 80 per km)	**1**
(v)	
$80x + 100 \leq 700 \Rightarrow 80x \leq 600 \Rightarrow x \leq 7.5$	**1**
Greatest whole number of kilometres = 7 km (check: 7 km costs Rs 660 $\leq$ 700; 8 km costs Rs 740 > 700)	**1**

Total: 10

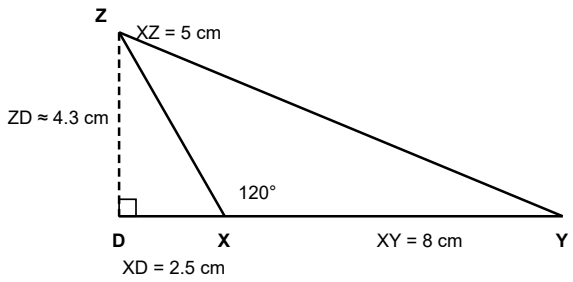
Question 4

(a)	
Multiply by 4: $(x + 3) + 2(x - 1) = 16$	**1**
$3x + 1 = 16 \Rightarrow 3x = 15$	**1**
$x = 5$ (check: $\frac{8}{4} + \frac{4}{2} = 4$)	**1**
(b) Let a = number of adult tickets, c = number of child tickets	
$a + c = 20$	**1**
$150a + 80c = 2300$ (equivalently $15a + 8c = 230$)	**1**
Eliminate or substitute: $150a + 80(20 - a) = 2300 \Rightarrow 70a = 700$	**1**
$a = 10$ adult tickets	**1**
$c = 20 - 10 = 10$ child tickets (check: $1500 + 800 = 2300$)	**1**
(c)	
$3n - 5 \leq 25$ (n = the whole number)	**1**
$n \leq 10 \Rightarrow$ the largest number is 10	**1**

Total: 10

Question 5

(i)	
XY = 8 cm drawn accurately with the ruler (accept 7.9 – 8.1 cm)	**1**
(ii) 120° at X (for example, 60° constructed twice from X, or 180° − 60°)	
Arc drawn from X cutting the line (or its extension), with the same compass setting kept	**1**
Arcs of equal radius used to obtain a 60° point, and then a second 60° point (or the 60° supplementary angle) with construction arcs shown	**1**
Arm drawn through the 120° point; angle 120° (accept ±2°)	**1**
(iii)	
Z marked with XZ = 5 cm (accept 4.9 – 5.1 cm) and triangle XYZ completed by joining ZY	**1**
(iv) Perpendicular from Z to YX produced	
YX produced beyond X	**1**
Perpendicular from an external point constructed with compasses (arcs from Z cutting the produced line at two points, then equal arcs from these points meeting)	**1**
Perpendicular ZD drawn and the foot named D	**1**
(v) ∠ZXD = 180° − 120° = 60°, so ZD = 5 sin 60° and XD = 5 cos 60°	
ZD ≈ 4.3 cm (theoretical 4.33 cm; accept 4.2 – 4.4 cm)	**1**
XD = 2.5 cm (accept 2.4 – 2.6 cm)	**1**



The finished figure (construction arcs not shown)

Total: 10

Question 6

(i)	
Area of trapezium = $\frac{1}{2} \times (20 + 14) \times 8$	**1**
= 136 m^2	**1**
(ii)	
$\pi r^2 = \frac{22}{7} \times 3.5 \times 3.5$	**1**
= 38.5 m^2	**1**
(iii)	
Area to be grassed = 136 − 38.5	**1**
= 97.5 m^2	**1**
Cost = 97.5 × 80 = Rs 7 800	**1**
(iv)	
Drop a perpendicular from C to AB: horizontal side = 20 − 14 = 6 m, vertical side = 8 m ; $BC^2 = 6^2 + 8^2 = 100$	**1**
BC = 10 m	**1**
(v)	
Perimeter = 20 + 10 + 14 + 8 = 52 m	**1**

Total: 10

Question 7

(a) Total number of cards = 20	
(i) Multiples of 5: 5, 10, 15, 20 ⇒ P = $\frac{4}{20} = \frac{1}{5}$	**1**
(ii) Square numbers: 1, 4, 9, 16 ⇒ P = $\frac{4}{20} = \frac{1}{5}$	**1**
(iii) Greater than 15: 16, 17, 18, 19, 20 ⇒ P = $\frac{5}{20} = \frac{1}{4}$	**1**
(iv) Not a multiple of 5: 20 − 4 = 16 ⇒ P = $\frac{16}{20} = \frac{4}{5}$ $(= 1 - \frac{1}{5})$	**1**
(b)	
(i) Sum = mean × number of values = 12 × 5	**1**
= 60	**1**
(ii) Sum of the four numbers = 10 + 15 + 8 + 14 = 47	**1**
Fifth number = 60 − 47 = 13	**1**
(iii) New sum = 60 + 18 = 78	**1**
New mean = 78 ÷ 6 = 13	**1**

Total: 10